

From Open Cholecystectomy to Laparoscopic Cholecystectomy: The University of Benin Teaching Hospital Experience

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ABSTRACT

The University of Benin Teaching Hospital (UBTH) general surgical laparoscopic surgery began in November 2020. It is one of the few tertiary institutions persistently offering laparoscopic services in south-south Nigeria. There has been remarkable increase in the number of cholecystectomies via laparoscopy over the years compared to the preceding 5 years of its commencement. This study aimed to highlight the remarkable change in the uptake of cholecystectomy in UBTH since the commencement of laparoscopic surgery. The records of cholecystectomies done over 10 years were reviewed; 5-years prior to and 5-years post-commencement of general surgery laparoscopy. All data obtained were analyzed using Excel. There were 28 cases of open cholecystectomies in 5 years prior to November 2020 (approximately 6 cases/year). The total number of cholecystectomies after commencing laparoscopic cholecystectomy was 135 over a 5-year period which included 17 cases of open cholecystectomy; approximately 3 cases/year (12.6%). There was a decline in the number of open cholecystectomies after commencing laparoscopy but it was not statistically significant (p value – 0.21). Laparoscopic cholecystectomies done over 5-years of its commencement is 118 cases; approximately 24 cases/year (87.4%). Each year witnessed a steady rise in the number of cases, which peaked in the 5th year. There was significant difference (p -value- 0.02) in the number of laparoscopic cholecystectomies compared to open cholecystectomies with the introduction of laparoscopic surgery in UBTH. The advent of laparoscopic general surgery has resulted in increased uptake of laparoscopic cholecystectomy in the UBTH. The authors advocate more investment in human and material resources in the ever-expanding field of minimal access surgery/laparoscopic surgery.

Keywords: Cholecystectomy, Open cholecystectomy, Laparoscopic cholecystectomy, South-South Nigeria.

INTRODUCTION

Cholecystectomy is the surgical removal of the gallbladder for gall bladder diseases which can be done open or via laparoscopy. Open cholecystectomy is the preferred method in Low Medium Income Countries (LMIC) though laparoscopic cholecystectomy is the gold standard. The disparity is largely due to capital intensive

nature of laparoscopic services and relative lack of skilled manpower to carry out laparoscopic procedures. The equipment, power supply and the training of surgeons are the core issues in the development of laparoscopy services in our locale.¹

Laparoscopic surgery has its clear advantages over open surgery. It is minimally invasive, has less post-operative pain, faster recovery from surgery, short hospital stays, early return to work, and more cosmetically acceptable.^{2,3,4} The cost of the procedure, availability of consumables for laparoscopic surgery and the lack of appropriate skills seem to be the major disadvantages in LMIC.¹ Open

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surgery is cheap, its skills ease to acquire, may take a shorter time to execute but it is more invasive, the scars may be cosmetically unacceptable, and prone to more respiratory complications, there is more post-operative pain, prolonged hospital stay, late return to work and more surgical site infection complications.^{5,6}

The transition from open cholecystectomy to laparoscopic cholecystectomy requires the training of surgeons to acquire the skills of laparoscopic surgery, training of support staff and the acquisition of laparoscopic equipment. This process involves time, money and the commitment of the surgical team with the support of hospital management. Factors which ultimately reinforce this process are the outcome of the procedures done and the continuous support of the hospital administration.⁷

The University of Benin Teaching Hospital (UBTH) general surgical laparoscopic surgery began 5 years ago. It is one of the very few tertiary institutions persistently offering laparoscopic surgery services in south-south Nigeria. A team of surgeons underwent training at the Lagos University Teaching Hospital (LUTH) to acquire the laparoscopic skills. This training was in addition to courses/training earlier attended by the surgeons. At the inaugural session of laparoscopic general surgery procedures in UBTH, in November 2020, a preceptor from LUTH was in attendance to give support to the team of UBTH surgeons. After the initial guidance, the UBTH surgeons were able to carry out the procedure independently, which provided opportunity for training and publications in scientific journals.^{8,9,10} There has been remarkable increase in the number of cholecystectomies via laparoscopy over the years compared to the preceding 5 years of its commencement.

This study aims to highlight the remarkable change in the uptake of cholecystectomy since the commencement of

laparoscopic surgery at the University of Benin Teaching Hospital.

PATIENTS/MATERIAL/METHODS

This was a retrospective descriptive observational study of all patients who underwent cholecystectomy between November 2015 and October 2025. The records of cholecystectomies done over 10 years (November 2015 – October 2025) were reviewed; 5 years prior to and 5 years post-commencement of general surgery laparoscopy. The number of cholecystectomies per year for open and laparoscopic cholecystectomies done over the years under review was entered into a proforma. The data points were sex, age, indication for surgery and outcome of surgery.

No identifiable personal health information was used for this study and the World Medical Association Helsinki protocol was adhered to.

All data obtained were entered into excel spreadsheet and then analyzed using Microsoft Excel 2019 version analysis tools; descriptive statistics and t-Test (Paired two samples for Means)

RESULTS

The number of open cholecystectomies over 5 years, prior to November 2020, was 28 cases (approximately 6 cases/year). There were 4 male (14.3%) and 24 female (85.7%) with ratio of 1:6. The mean age was 47.1+/-14.2 years and age range of 22-81 years.

The total number of cholecystectomies after commencing laparoscopic cholecystectomy was 135 over a 5-year period which included 17 cases of open cholecystectomy; approximately 3 cases/year (12.6%) (Table 2 and Figure 1). For the open cholecystectomy, there were 1 male (4.5%) and 21 female (95.4%) with ratio of 1:21. The mean age was 48.2+/-14.1 years and age range of 11-66 years. There was a decline in the number of open cholecystectomies after commencing

laparoscopic cholecystectomy but it was not statistically significant (p value – 0.21) (Figure 2 and Figure 3).

Table 1: Indications for cholecystectomy over the ten-year study period in UBTH

INDICATIONS	NUMBER	PERCENTAGE
Symptomatic cholelithiasis	154	91.7
Choledocholithiasis	9	5.4
Gallbladder polyp	2	1.2
Porcelain gallbladder	1	0.6
Gallbladder Empyema	1	0.6
Acalculous cholecystitis	1	0.6

Laparoscopic cholecystectomy cases done over 5 years of its introduction in UBTH was 118 cases; approximately 24 cases/year (87.4%). There were 14 male (11.9%) and 104 female (88.1%) with ratio of 1:6. The mean age was 44.6+/-15.8years and age range of 20-82years. Each year witnessed a steady rise in the number of cases with the 5th year having the highest number of cases (Table 2 and Figure 2

There was a significant difference (p-value- 0.02) in the number of cholecystectomies done by laparoscopy compared to those done open with the introduction of laparoscopic surgery in UBTH.

The main indication for cholecystectomy in our study is symptomatic cholelithiasis which constituted 97.1%. The other indications are shown in Table 1.

The complications of cholecystectomy were surgical site infection (SSI) and biloma. Of the 50 patients that had open cholecystectomy over the 10-year period, 11patients (22%) had SSI while 6 patients (5%) had port site infection following laparoscopic cholecystectomy in 118 patients. Only one patient had biloma following laparoscopic cholecystectomy and it was treated by percutaneous drainage. There was no mortality.

Table 2: Yearly number of cholecystectomies after introduction of Laparoscopic cholecystectomy in UBTH

Year	Number of Open Cholecystectomy	Number of Laparoscopic Cholecystectomy
2021	5	6
2022	2	20
2023	2	29
2024	4	30
2024	4	33

DISCUSSION

There were predominantly females in our study in their 5th decade of life and the main indication for cholecystectomy was symptomatic cholelithiasis. This finding is in tandem with studies done in other centres within Nigeria^{11,12,13}. Our study also demonstrated that Laparoscopic cholecystectomy has low surgical site infection rate compared to open cholecystectomy and has fewer complications, a well-established advantage of laparoscopic cholecystectomy.^{5, 6}

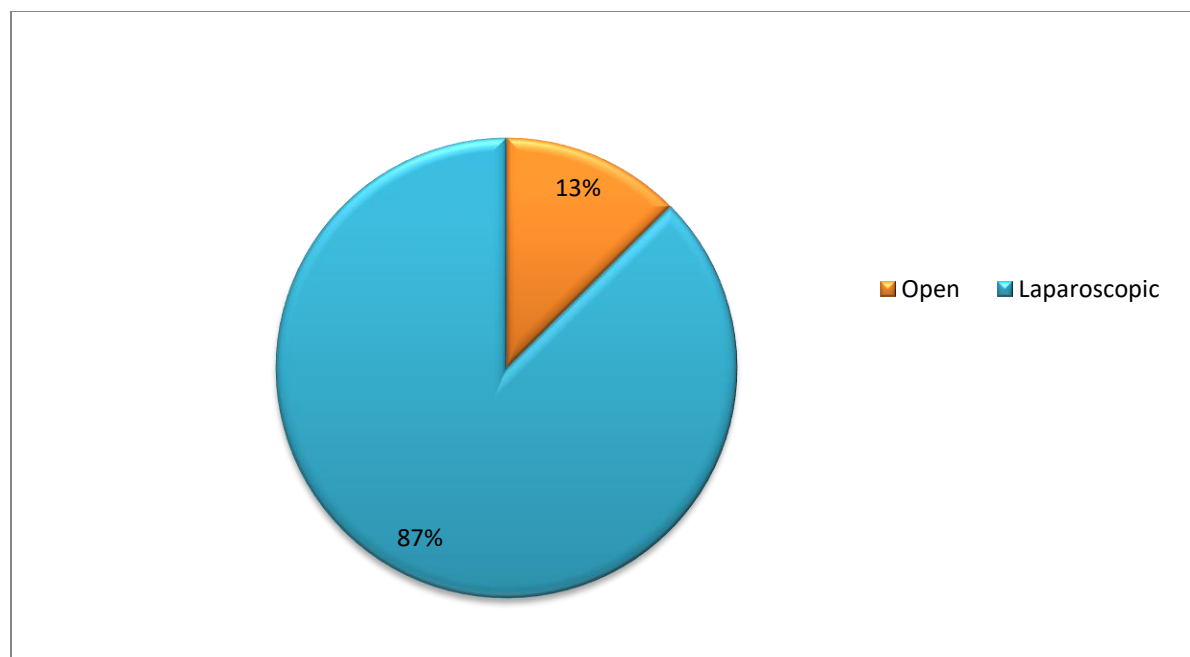


Figure 1: Type of cholecystectomies 5 years after introduction of laparoscopic cholecystectomy in UBTH

Our study shows a remarkable increase in the number of cholecystectomies done following the introduction of laparoscopic cholecystectomy with a peak in the fifth year while open cholecystectomy is on the decline (Figures 2 and 3).

Studies have shown increased uptake of cholecystectomy with the introduction of laparoscopy.^{14, 15, 16} The reasons for this increase were ascribed to the broadening of the indication for cholecystectomy and a large pool of patients with mild symptoms that had their surgery deferred till the advent of laparoscopic cholecystectomy.¹⁶ In our study, this may have been due to advocacy for the procedure at inception, short hospital stay and improved cosmesis.

The steady rise seen in the number of cases over the 5-year period post-

commencement of laparoscopic cholecystectomy (Table 2 and Figure 2), may be due to gradual acceptability of the procedure and enhanced technical proficiency of the surgeons. This finding is similar to the study in Ife and elsewhere at the inception of laparoscopic cholecystectomy.¹¹

There was a reduction in the number of open cholecystectomies and a disproportionate increase in the number of laparoscopic cholecystectomies in our study (Figure 2). This may imply that the population of patients for laparoscopic cholecystectomy may have different demographic characteristics which need to be explored in future studies and the acceptability of the procedure. Studies have shown similar trends in centres in the early years of laparoscopic cholecystectomy.^{12, 13, 17, 18}

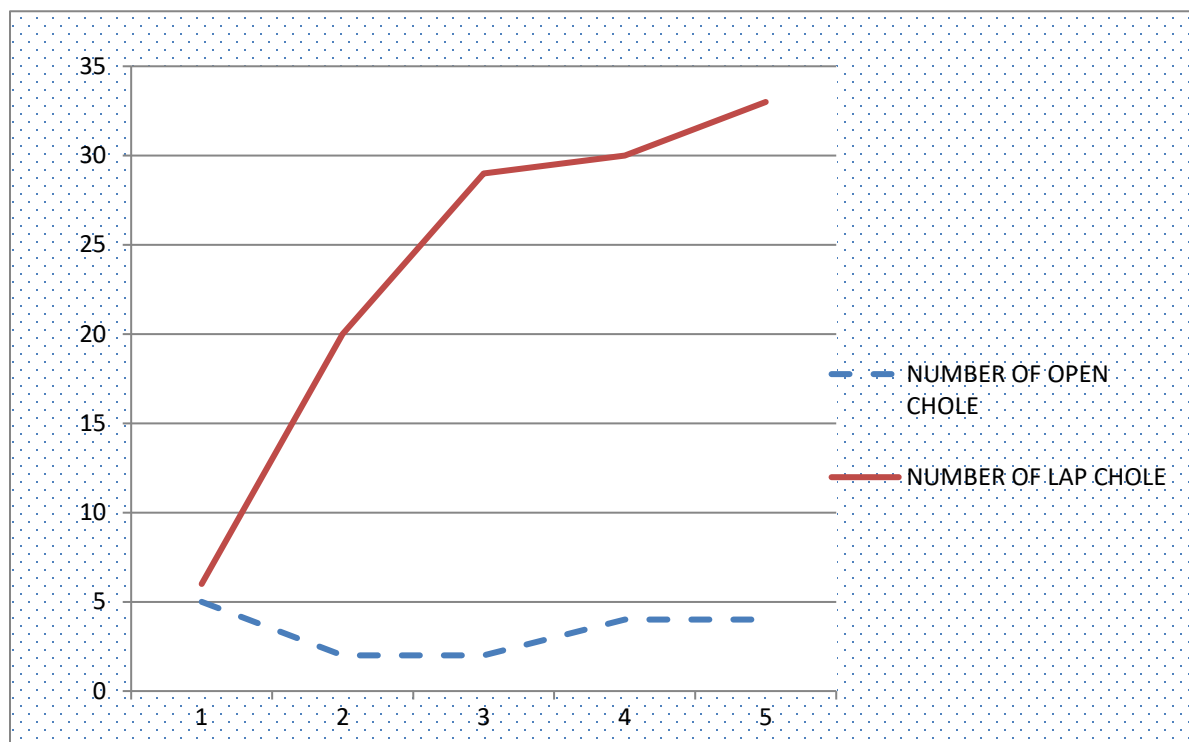


Figure 2: Trends of open and laparoscopic cholecystectomy after introduction of laparoscopic cholecystectomy in UBTH

Our study has implications for massive investment in human and material resources and training because for every patient in whom cholecystectomy is indicated, there is near 90% chance (Figure 1) that they will have laparoscopic cholecystectomy.^{19,20,7}

The apparent low uptake of cholecystectomy in the pre-laparoscopy years (Figure 3) in our centre is largely due to non-availability of laparoscopic services at the time and probably the fear of complications associated with open cholecystectomy. The introduction of this service in our centre; being a tertiary institution, led to a surge in the procedure due to referral. Similar surge in cholecystectomy was also found in Abuja, Ife and Jos with the introduction of laparoscopy^{11, 12, 13}.

Though our study did not consider the cost, which remains an important factor in

surgical interventions in LMIC, it demonstrated reasonable measure of acceptability of laparoscopic cholecystectomy despite the cost of procedure.^{21,22}

The transition from open cholecystectomy to laparoscopic cholecystectomy is usually not an easy venture in institutions where the open procedure is well established. The training of skilled manpower takes some time, and the purchase and maintenance of equipment may also be challenging. What makes the difference is the commitment of the surgical team to make this transition a reality. Successes in the initial procedures, as well as the rapidly evolving skills of the surgical team serve as positive feedback for the development of laparoscopic cholecystectomy as well as other laparoscopic procedures.⁷

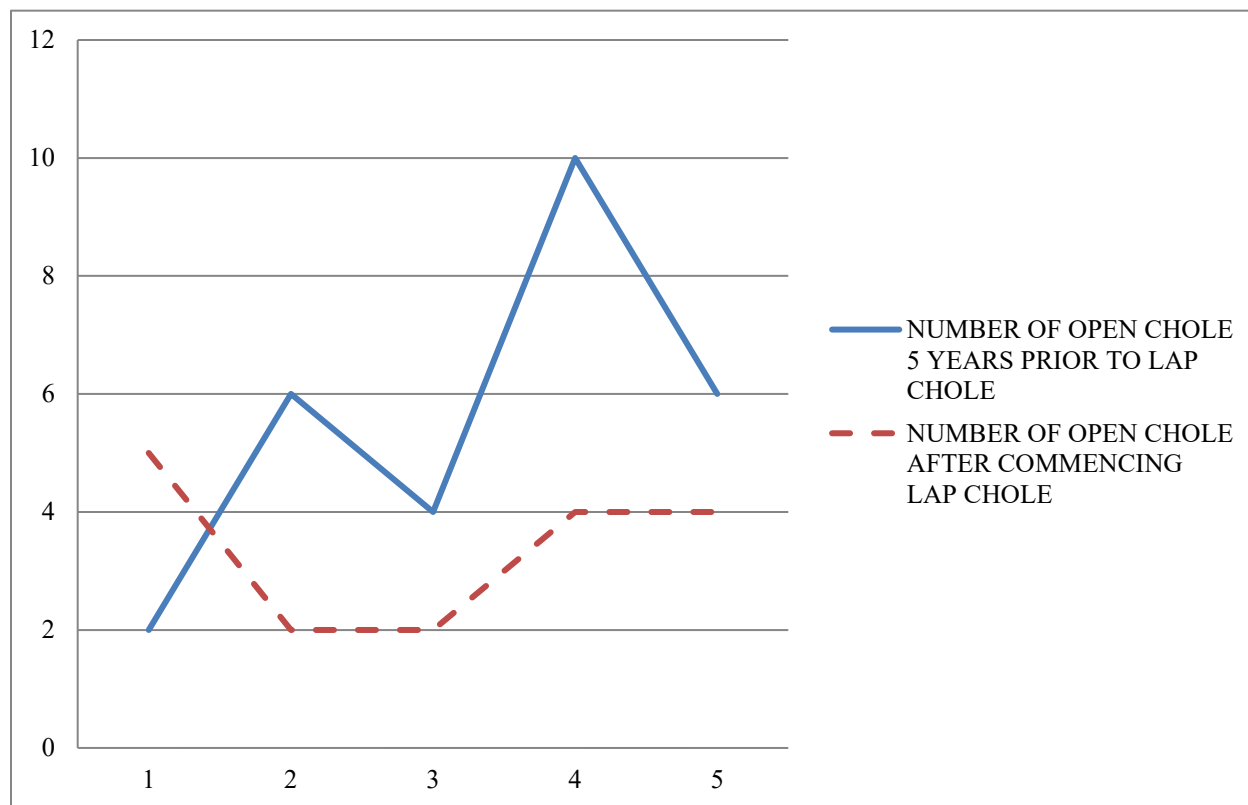


Figure 3: Comparing open cholecystectomy before and after introduction of laparoscopic cholecystectomy

The increasing number of laparoscopic cholecystectomies performed in our centre will open up opportunity for training of resident doctors and research. This is evidenced by the publications that have emanated from laparoscopic general surgery since its commencement.^{8,9,10} We have demonstrated in our study that introduction of laparoscopic surgery resulted in the increased uptake of cholecystectomy. The transition from open to laparoscopic surgery requires training of manpower and subsequent dedication of the surgical team

Conclusion

The introduction of laparoscopic services was associated with increase in the uptake of laparoscopic cholecystectomy in the UBTH. The authors advocate more investment in human and material resources in ever-expanding field of minimal access

surgery/laparoscopic surgery and recommend future multicentre studies.

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